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Section A : Write the OUTPUT of the following Code

Note :

- 1. In case your answer is error – specify the reason for error.**
- 2. Assume that all necessary header files are already included.**
- 3. Write your answer in the given table.**

Q#	1.	2.	3.	4.	5.
Answer	4 8 12 16 20	7 7 4 9 6 11 8 13 10 15	2 2 2 2 2 2 3 4 6 5	Garbage	16 16 16
Q#	6.	7.	8.	9.	10.
Answer	ASCII equivalent to number 1	10	-1 1	1 10	7 6 5 4 3

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1. <code>int main()</code> [2 Marks] <pre>{ int a[] = { 2, 4, 6, 8, 10 }; int i; for (int i = 0; i <= 4; i++) { *(a + i) = a[i] + i[a]; cout << *(i + a) << endl; } return 0; }</pre>	Rough Work:
2. <code>void f(int *x, int *y)</code> [2 Marks] <pre>{ *x = *(y) += 2; } int main() { int a[5] = { 2, 4, 6, 8, 10 }; int i,b=5; for (int i = 0; i <= 4; i++) { f(a, &b); cout << a[i] << " " << b << endl; } return 0;} }</pre>	Rough Work:
3. <code>int main()</code> [2 Marks] <pre>{ int c[] = { 2.8, 3.4, 4, 6.7, 5 }; int j, *p = c, *q = c; for (j = 0; j < 5; j++) { cout << *c << endl; ++q; } for (j = 0; j < 5; j++) { cout << *p << endl; ++p; } return 0;} }</pre>	Rough Work:
4. <code>int main()</code> [2 Marks] <pre>{ int one_d[] = { 1, 2, 3 }; int *ptr; ptr = one_d; ptr += 3; cout << *ptr << endl; return 0; }</pre>	Rough Work:
5. <code>int main()</code> [2 Marks] <pre>{ int a = 2, *f1, *f2; f1 = f2 = &a; *f2 += *f2 += a += 2; cout << a << " " << *f1 << " " << *f2 << endl; return 0; }</pre>	Rough Work:

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}	
<pre> 6. int main() { char not; not = !(0); cout << not << endl; return 0; } </pre>	[1 Marks] <i>Rough Work:</i>
<pre> 7. int main() { int *j; { int i = 10; j = &i; } cout << *j << endl; return 0; } </pre>	[1 Mark] <i>Rough Work:</i>
<pre> 8. int main() { int i = -1; -i; cout << i << " " << -i << endl; return 0; } </pre>	[1 Mark] <i>Rough Work:</i>
<pre> 9. int main() { int i = 5, j = 10; i = i &= j && 10; cout << i << " " << j << endl; return 0; } </pre>	[1 Mark] <i>Rough Work:</i>
<pre> 10. void change(int *b) { int i; for (i = 0; i <= 4; i++) { *b = *b + 1; b++; } } int main() { int a[5] = { 2, 3,4,5,6 }; int i; change(a); for (int i = 4; i >=0; i--) { cout << a[i] << endl; } return 0; } </pre>	[3 Marks] <i>Rough Work:</i>

SECTION B

Note: Choose appropriate data types for the arguments and return type, and use constant (const), wherever required.

1. Write a program to determine whether given number is **Harshad Number(or Niven Number)** or not:

[Note: If a number is divisible by the sum of its digits then it will be known as a Harshad Number.

Sample Output:

1. Input Number: 156
Sum of Digit = $1+5+6=12$
 $156/12=13$
156 is a Harshad (Niven) Number
2. Input Number: 23
Sum of Digit= $2+3=5$
23 is not divisible by 5
23 is not Harshad (Niven) Number

```
#include<iostream>
using namespace std;
int main()
{
    int n, num, sum = 0, rem;
    cout << "Input number: ";
    cin >> n;
    num = n;
    while (n != 0)
    {
        rem = n % 10;
        n = n / 10;
        sum = sum + rem;
    }

    cout << "Sum of digits = " << sum << endl;
    if ((num%sum) == 0)
    {
        cout << num << "is Harshad(Niven) number" << endl;
    }
    else
    {
        cout << num << "is not Harshad(Niven) number" << endl;
    }
    system("pause");
    return 0;
}
```

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2. Given two integer arrays $a1[n+1]$ and $a2[n]$ i.e. number of elements in array $a1$ is one more than the array $a2$. Elements of $a2$ are the subset of $a1$, (i.e. $a2$ contains all values from $a1$ except one, and values in $a2$ can be in any order).

Write a program to find the element which is present in $a1$ but not in $a2$.

[Sample Output:

1. Input $a1$: 1,2,3,4,5
Input $a2$: 3,4,1,5
Missing element= 2

2. Input $a1$: 2,8,9,7,6
Input $a2$: 6,9,8,2
Missing Element=7

```
#include<iostream>
using namespace std;
int main()
{
    int a[15], b[14], n,sum=0,sum1=0;
    cout << "Input number of elements in array a: ";
    cin >> n;
    cout << "Input a: ";
    for (int i = 0; i < n; i++)
    {
        cin >> a[i];
    }
    cout << "*****" << endl;
    for (int i = 0; i < n; i++)
    {
        cout << a[i] << " ";
    }
    cout << endl;
    cout << "Input array b: ";
    for (int i = 0; i < n-1; i++)
    {
        cin >> b[i];
    }
    cout << "*****" << endl;
    for (int i = 0; i < n-1; i++)
    {
        cout << b[i] << " ";
    }
    cout << endl;
    for (int i = 0; i < n; i++)
    {
        sum += a[i];
    }
    for (int i = 0; i < n-1; i++)
    {
        sum1 += b[i];
    }
    cout << "Missing Number in array b is = " << sum - sum1 << endl;
    system("pause");
    return 0;
}
```